2d. Content of Statistics (Optional paper Y542)

Introduction to Statistics.

In **Statistics** learners will explore the theory which underlies the statistics content in A Level Mathematics, as well as extending their tool box of statistical concepts and techniques. This area covers probability involving combinatorics, probability distributions for discrete and continuous random variables, hypothesis tests and confidence intervals for a population mean, chi-squared tests, non-parametric tests, correlation and regression.

5.01 Probability
The work on probability in A Level Mathematics is extended to include problems involving arrangements and selections.

5.02 Discrete Random Variables
The general concept of a discrete random variable introduced in A Level Mathematics is further developed, along with the calculation of expectation and variance. The discrete uniform, binomial, geometric and Poisson distributions are studied.

5.03 Continuous Random Variables
The general concept of a continuous random variable introduced in A Level Mathematics is further developed, including probability density functions and cumulative distribution functions. Calculus is used to find expectation, median and quartiles.

5.04 Linear Combinations of Random Variables
Formulae are introduced and applied for linear combinations.

5.05 Hypothesis Tests and Confidence Intervals
The study of hypothesis tests is formalised and developed further, including the central limit theorem. The concept of a confidence interval is introduced, applied in context.

5.06 Chi-squared Tests

5.07 Non-parametric Tests
The concept of a non-parametric test is introduced and explored through the Wilcoxon signed-rank tests and the Wilcoxon rank-sum test (or Mann-Whitney U test), and applied in hypothesis tests concerning population median and identity of populations.

5.08 Correlation
The concept of correlation introduced in A Level Mathematics is formalised and explored further, including the study of rank correlation.

5.09 Linear Regression
Regressions lines are calculated and used in context for estimation.

Assumed knowledge
Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but will only be explicitly assessed where it appears in this section.

Use of technology

To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. Learners should use spreadsheets or statistical software to generate tables and diagrams, and to perform standard statistical calculations.

2. Hypothesis tests: Learners should use spreadsheets or statistical software to carry out hypothesis tests using the techniques in this paper.

3. Probability: Learners should use random number generators, including spreadsheets, to simulate tossing coins, rolling dice etc, and to investigate probability distributions.

5 Use of data: Learners are expected to have explored different data sets, using appropriate technology, during the course. No particular data set is expected to be studied, and there will not be any pre-release data.

Hypothesis Tests

Hypotheses should be stated in terms of parameter values (where relevant) and the meanings of symbols should be stated. For example,
" $H_0: p = 0.7, H_1: p < 0.7$, where p is the population proportion in favour of the resolution".

Conclusions should be stated in such a way as to reflect the fact that they are not certain. For example,
"There is evidence at the 5% level to reject H_0. It is likely that the mean mass is less than 500 g."
"There is no evidence at the 2% level to reject H_0. There is no reason to suppose that the mean journey time has changed."

Some examples of incorrect conclusion are as follows:
" H_0 is rejected. Waiting times have increased."
"Accept H_0. Plants in this area have the same height as plants in other areas."

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When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
5.01 Probability			
5.01a	Probability	a) Be able to evaluate probabilities by calculation using permutations and combinations. *Includes the terms "permutation" and "combination".* *Includes the notation* ${}_n\mathrm{P}_r = {}^n\mathrm{P}_r$ *and* ${}_n\mathrm{C}_r = {}^n\mathrm{C}_r$. *For underlying content on probability see H240 section 2.03.*	
5.01b		b) Be able to evaluate probabilities by calculation in contexts involving selections and arrangements. *Selection problems include, for example, finding the probability that 3 vowels and 2 consonants are chosen when 5 letters are chosen at random from the word 'CALCULATOR'.* *Arrangement problems only involve arrangement of objects in a line and include:* *1. repetition, e.g. the probability that the word 'ARTIST' is formed when the letters of the word 'STRAIT' are chosen at random.* *2. restriction, e.g. the probability that two consonants are (or are not) next to each other when the letters of the word 'TRAITS' are placed in a random order.*	